Musings about linear algebra

What is linear algebra good for? Some snippets from Quora answers:

"I didn't really know what linear math was good for. Then, I decided to write 3D computer games from the ground up. I discovered I needed my linear algebra notes."

PageRank, probability-statistics, control theory, economics, AI (machine learning, LLM) . . .

In many interesting places, after getting to a certain step, it's "just linear algebra from there".

(Linear algebra is) *"the Swiss Army knife of math", "a gateway drug into higher math."*

A little more technically, linear algebra (among other things) gives systematic ways to solve and think about systems of linear equations in many variables, a situation that occurs frequently. Fortunately linear systems are relatively easy to understand *and even in nonlinear situations*, many times we can understand a lot using linear algebra together with calculus.

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0. What is linear algebra? Like the blind men describing an elephant, there can be many valid answers. Here is one, with a nod to the gateway answer quoted above[1].

The foundational major concept is that of a **linear map**[2] *(or transformation or function).* If you know the simple algebraic definition, good for you. But also possibly bad, because the simplicity is likely unearned and hence un(der)appreciated. Let's try to make amends.

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1. How *should* we define a *linear* function? The name suggests a geometric provenance, so we might call a function linear if its *graph* is a straight line. This answer presupposes that both the domain and the codomain are the real line $\mathbb{R}$. What if the domain is, say, the Euclidean plane $\mathbb{R}^2$? No matter what the codomain is, the "graph" cannot then be a line. So we have to find a different way. Here is a natural attempt at a definition.

Definition. Let us call a function f line-preserving if f(any straight line) = a straight line. (We use a different name because this notion is not quite standard.)

Why is this definition natural? You have the right to disagree, but as you learn more mathematics, hopefully you will come to see that in many contexts it is highly worthwhile to study "functions preserving . . . ", where ". . . " stands for some structure of interest. Be that as it may, let's accept the given definition and get to some problems.

(a) Try to describe[3] all line-preserving functions from $\mathbb{R}$ to $\mathbb{R}$, from $\mathbb{R}$ to $\mathbb{R}^2$, from $\mathbb{R}^2$ to $\mathbb{R}$, and from $\mathbb{R}^2$ to $\mathbb{R}^2$. What if we relax the definition to f(any straight line) $\subset$ a straight line? Are all subparts equally interesting?

(b) Define and describe all distance-preserving functions from $\mathbb{R}^2$ to $\mathbb{R}^2$. Do the same for angle-preserving functions. What about such functions from $\mathbb{R}^3$ to $\mathbb{R}^3$? From $\mathbb{R}^3$ to $\mathbb{R}^2$?

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[1] We will give other answers later, e.g., see item 3 on the next page.
[2] A vector space is "just" something we will define to make sense of what a linear map is.
[3] What is an acceptable description? Well, a simpler one. What does simpler mean? A valid question, but here let me rest by quoting the famous "I know it when I see it" quip by Potter Stewart.

2. Rotations. The composition of two rotations of $\mathbb{R}^2$ around the origin is clearly a rotation. Is the same true in $\mathbb{R}^3$? Here a rotation is around an *axis*, which we take to be through the origin. It is a marvellous fact that the composition of two such rotations is also a rotation.

We'll prove this with linear *algebra*. Can you do it geometrically? Any further questions that are almost begging to be asked? Any connection with something in the previous problem?

An example of linear algebra as a gateway. A great gift of higher math is a vastly enlarged and multifaceted conception of geometry. Building on linear algebra, down the line one sees that the *space of all rotations* is also a geometric object. Of what sort? A mathematically rigorous understanding needs some group theory and topology, but you can look up and try to decode a couple of fun things now: "How about sending me a fourth gimbal for Christmas?" and Dirac's belt trick, e.g., this quick demonstration (in a cool accent too).

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3. Need for new words. (a) What are the possible ways in which a line L and a line M in the plane can intersect? Answer: the intersection is empty if the lines are parallel, exactly one point when they are distinct and non-parallel, and an entire line if L = M. Algebraically, this translates to describing the solution set of a system two linear equations in two variables. Try to describe the three cases purely algebraically, with no reference to geometry.

Now try to do this exercise when the number of equations and number of variables both increase. What can happen when we have 5 linear equations in 7 variables? 7 equations in 5 variables? 5 equations in 5 variables? Does the trichotomy of 0/1/infinitely many solutions continue to hold? Is this a good enough description of the solution set?

Linear algebra gives new language needed to even talk sensibly about the possibilities. We need further language to find all solutions, a language that deals in algebra but is strongly informed by geometric intuition. (This marriage of algebra and geometry is a major theme in mathematics.) *Words involved: subspace, dimension, basis, row reduction, pivots, etc.*

(b) You know the formula for Fibonacci numbers, the one in terms of $\sqrt{5}$. You can prove it with induction. How do you *find* the formula? You may know that the polynomial $x^2 - x - 1$ has something to do it and possibly that there is a characteristic equation associated with any homogeneous recurrence relation. But how does one understand why any method like this *should be there* in the first place? To quote the great William Thurston, "The product of mathematics is clarity and understanding. Not theorems, by themselves."[4]

We'll see that the Fibonacci situation, and many others, can be understood in terms of a good basis for a vector space. In fact **much of linear algebra is search for a good basis**.

For example, we will see that every symmetric matrix with real entries can be "*diagonalized*" in a nice way (after we see the meaning of diagonalization and the nice way). This theorem has myriad applications, e.g., in optimization of a multivariable function by generalizing the second derivative test (to be seen next semester), analysis of covariance matrix in statistics, physics (quantum mechanics), image compression, spectral graph theory, ...

Here is a related problem for you. Let A be an $n \times n$ matrix. Let xI be the $n \times n$ matrix with each diagonal entry = the variable x and all other entries 0. The determinant of $xI - A$, calculated using the usual rules, is a polynomial of degree n. For symmetric A with real entries, this polynomial has only real roots. See this for $n = 2$. Can you show it for $n = 3$?

[4]The full Thurston quote is here. In particular Thurston says "...mathematics only exists in a living community of mathematicians that spreads understanding ..." Let's try to build such a community!